

AI & ML – 40 Hours Internship Module

Module Duration

- Total Duration: 40 Hours
- Duration: 2 Weeks
- Daily Class: 4 Hours

Module Outline

Module 1: Introduction to AI & ML (4 Hours)

- Introduction to Artificial Intelligence
- Types of AI
- Introduction to Machine Learning
- Applications of AI & ML
- AI Industry Trends and Career Opportunities

Module 2: Python for AI & ML (6 Hours)

- Python Basics
- Variables, Data Types, Operators
- Conditional Statements & Loops
- Functions and Libraries
- NumPy & Pandas Basics

Module 3: Data Handling & Visualization (4 Hours)

- Data Collection Techniques
- Data Cleaning
- Handling Missing Values
- Data Visualization using Matplotlib & Seaborn
- Exploratory Data Analysis (EDA)

Module 4: Machine Learning Fundamentals (6 Hours)

- Supervised vs Unsupervised Learning
- Regression Algorithms
- Classification Algorithms
- Clustering Basics

- Model Training & Testing

Module 5: Machine Learning Algorithms (6 Hours)

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- K-Nearest Neighbors (KNN)

Module 6: Model Evaluation & Optimization (4 Hours)

- Accuracy, Precision, Recall
- Confusion Matrix
- Overfitting & Underfitting
- Cross Validation
- Hyperparameter Tuning Basics

Module 7: Introduction to Deep Learning (4 Hours)

- Neural Networks Basics
- TensorFlow & Keras Introduction
- ANN Concepts
- AI Model Demonstration

Module 8: AI Tools & Generative AI Basics (3 Hours)

- Introduction to Generative AI
- ChatGPT & AI Productivity Tools
- Prompt Engineering Basics
- AI Automation Use Cases

Module 9: Mini Project & Presentation (3 Hours)

- AI/ML Mini Project
- Dataset Implementation
- Model Deployment Overview
- Final Presentation & Evaluation

Deliverables

- Internship Completion Certificate
- Study Materials
- Hands-on Practice Sessions

- Mini Project
- Industry Exposure Sessions

Recommended Tools

- Python
- Jupyter Notebook
- Google Colab
- VS Code
- TensorFlow
- Scikit-learn